

SpaceX Falcon 9 First- Stage Landing Prediction

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<https://github.com/hodzz/IBM-SpaceX-Data-Science-Capstone>

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Executive Summary

- This project analyzed SpaceX Falcon 9 launch data to identify the factors associated with successful first-stage landings. Data was collected from the SpaceX API and a Wikipedia page, cleaned and prepared using Pandas, and explored using Python visualizations and SQL.
 - Interactive analysis was performed using Folium maps and a Plotly Dash dashboard. Four classification models were trained and compared: Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbors.
 - The models achieved similar test performance, with an accuracy of approximately 83.33%. The results also showed that landing success generally improved over time and varied according to launch site, orbit, payload mass, and flight experience.
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- Key Results
 - Landing success improved over time.
 - Launch site, orbit, payload mass, and flight experience influenced outcomes.
 - Four classification models achieved approximately 83.33% test accuracy.

Introduction and Business Problem



SpaceX Falcon 9 Rocket – The Verge

Background

SpaceX has reduced the cost of space launches by recovering and reusing the Falcon 9 first-stage booster.

A Falcon 9 launch costs approximately \$62 million, compared with much higher prices for some traditional launch providers.

The ability to recover the first stage is one of the main reasons SpaceX can offer lower launch prices.

Problem: The objective of this project is to build a machine-learning model that predicts whether the Falcon 9 first stage will land successfully.

A successful landing means the booster may be reused, which can reduce the overall cost of the mission.

Business Question:

Can Falcon 9 launch characteristics be used to predict whether the first-stage booster will land successfully?

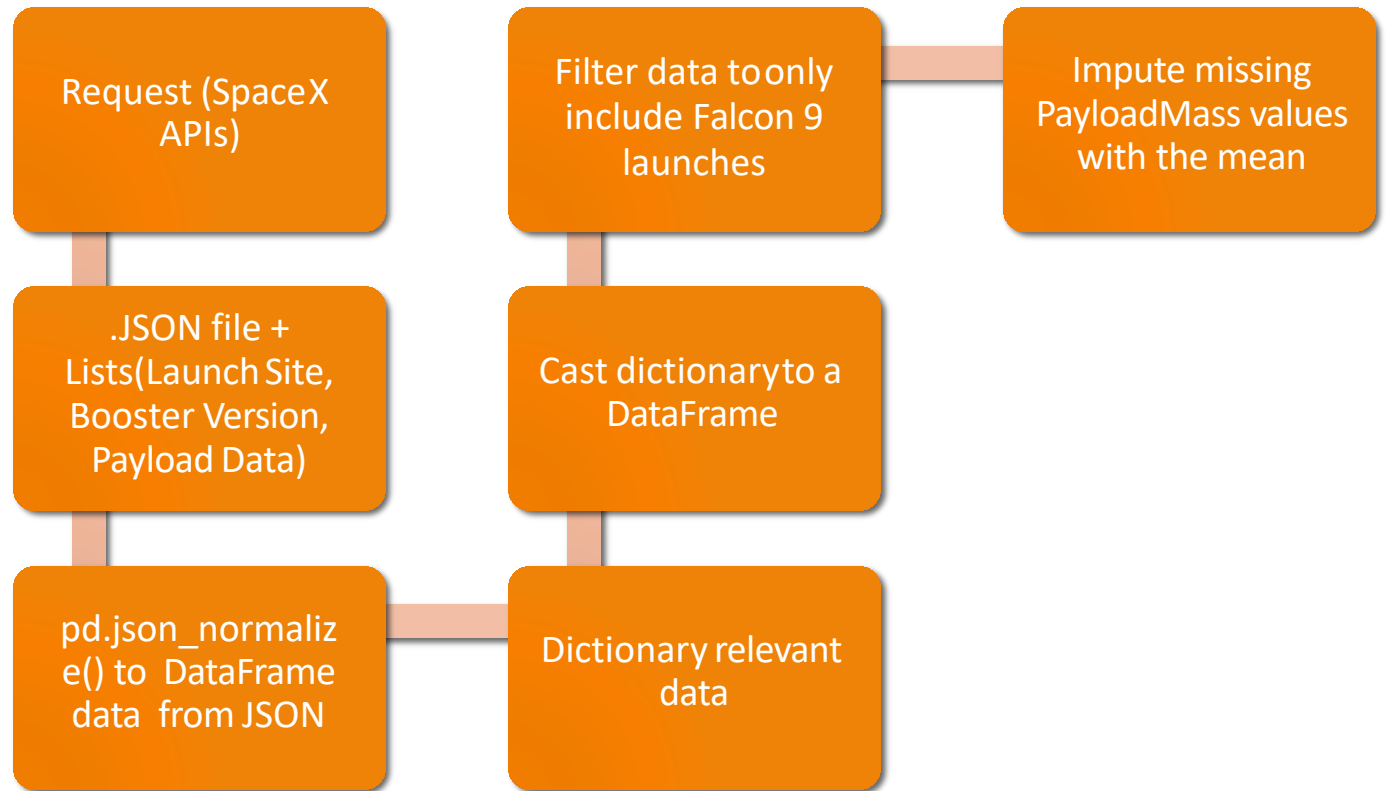
Project Methodology Overview

- 1. Collect Falcon 9 launch data using the SpaceX API.
- 2. Collect additional launch data through Wikipedia web scraping.
- 3. Clean the data and create a binary landing-success label.
- 4. Perform EDA using Python visualizations and SQL.
- 5. Explore launch-site geography using Folium.
- 6. Build a Plotly Dash dashboard.
- 7. Standardize and encode the features.
- 8. Train Logistic Regression, SVM, Decision Tree, and KNN models.
- 9. Compare test accuracy and confusion matrices.

Data Collection Methodology — SpaceX API

GitHub url:

<https://github.com/hodzz/IBM-SpaceX-Data-Science-Capstone/blob/main/Data%20Collection%20Api.ipynb>



Output:

A Falcon 9 launch DataFrame containing flight number, launch date, booster version, payload mass, orbit, launch site, landing outcome, and core information.

Data Collection Methodology — Web Scraping

GitHub url:

<https://github.com/hodzz/IBM-SpaceX-Data-Science-Capstone>

1. Request the Wikipedia Falcon 9 launch page.
2. Parse the HTML using BeautifulSoup and html5lib.
3. Identify launch tables and column headers.
4. Iterate through table rows and extract launch information.
5. Store the data in a dictionary.
6. Convert the dictionary into a Pandas DataFrame.

Output:

A structured DataFrame containing launch number, date, launch site, payload, orbit, customer, launch outcome, and booster landing result.

Data Wrangling Methodology

- Removed Falcon 1 launches so the analysis focused only on Falcon 9 missions.
- Checked the dataset for missing values and replaced missing PayloadMass values with the mean payload mass.
- Converted the Outcome column into a binary target variable called Class.

Class = 1: Successful landing

True ASDS, True RTLS, True Ocean

Class = 0: Unsuccessful landing

None None, False ASDS, None ASDS, False Ocean, False RTLS

- The cleaned dataset was then used for data visualization, SQL analysis, Folium mapping, Plotly Dash, and machine-learning classification.

Output: A cleaned Falcon 9 dataset with a binary landing-success label.

GitHub: <https://github.com/hodzz/IBM-SpaceX-Data-Science-Capstone>

EDA and Data Visualization Methodology

Variables analysed:

Flight Number, Payload Mass, Launch Site, Orbit, Class, and Year

Charts created:

- Flight Number vs Launch Site
- Payload Mass vs Launch Site
- Success Rate by Orbit
- Flight Number vs Orbit
- Payload Mass vs Orbit
- Yearly Landing-Success Trend

Purpose:

Scatter plots, bar charts, and line charts were used to identify relationships between launch characteristics and landing success.

GitHub url: <https://github.com/hodzz/IBM-SpaceX-Data-Science-Capstone>

EDA with SQL Methodology

- Loaded the SpaceX dataset into a SQLite database.
- Stored the data in the SPACEXTBL table.
- Used SQL magic commands inside Python.
- Applied SELECT, DISTINCT, WHERE, GROUP BY, ORDER BY, COUNT, SUM, AVG, MIN, and MAX.
- Analysed launch sites, payload masses, mission outcomes, booster versions, successful landings, rankings, and dates.

<https://github.com/hodzz/IBM-SpaceX-Data-Science-Capstone>

Interactive Visual Analytics Methodology — Folium

- Marked all launch sites on an interactive Folium map.
- Added green markers for successful landings and red markers for unsuccessful landings.
- Used MarkerCluster to group launches with similar coordinates.
- Calculated distances from launch sites to coastlines, railways, highways, and cities.
- Used the map to examine launch-site safety and logistical positioning.

<https://github.com/hodzz/IBM-SpaceX-Data-Science-Capstone>

Build a Dashboard with Plotly Dash

Dashboard includes a pie chart and a scatter plot.

Pie chart can be selected to show distribution of successful landings across all launch sites and can be selected to show individual launch site success rates.

Scatter plot takes two inputs: All sites or individual site and payload mass on a slider between 0 and 10000 kg.

The pie chart is used to visualize launch site success rate.

The scatter plot can help us see how success varies across launch sites, payload mass, and booster version category.

GitHub url:

<https://github.com/hodzz/IBM-SpaceX-Data-Science-Capstone>

Predictive analysis Methodology

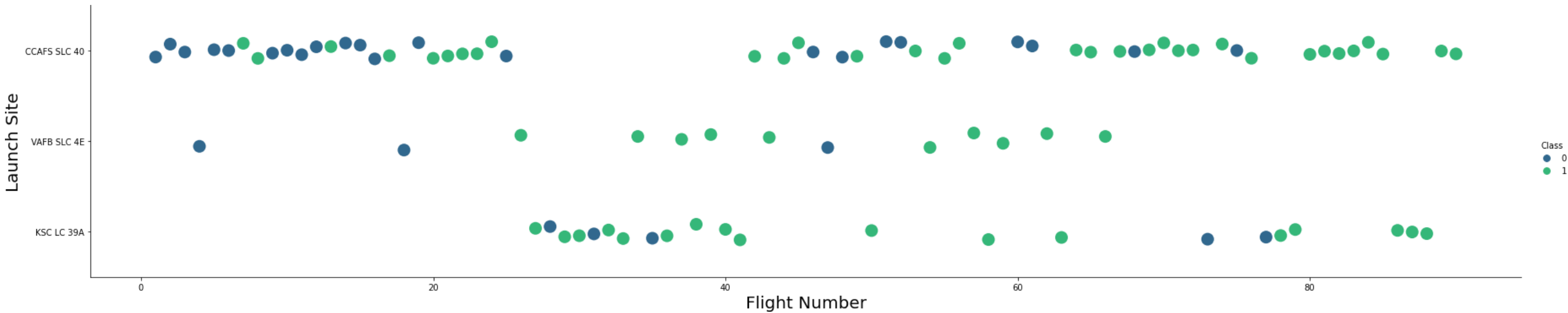
GitHub url:

<https://github.com/hodzz/IBM-SpaceX-Data-Science-Capstone>

1. Separate the Class target from the feature dataset.
2. Apply one-hot encoding to categorical variables.
3. Standardize the numerical features using StandardScaler.
4. Split the dataset into training and test samples.
5. Use GridSearchCV with cross-validation.
6. Train Logistic Regression, SVM, Decision Tree, and KNN.
7. Compare validation and test accuracy.
8. Review confusion matrices.

Test-set size: 18 records

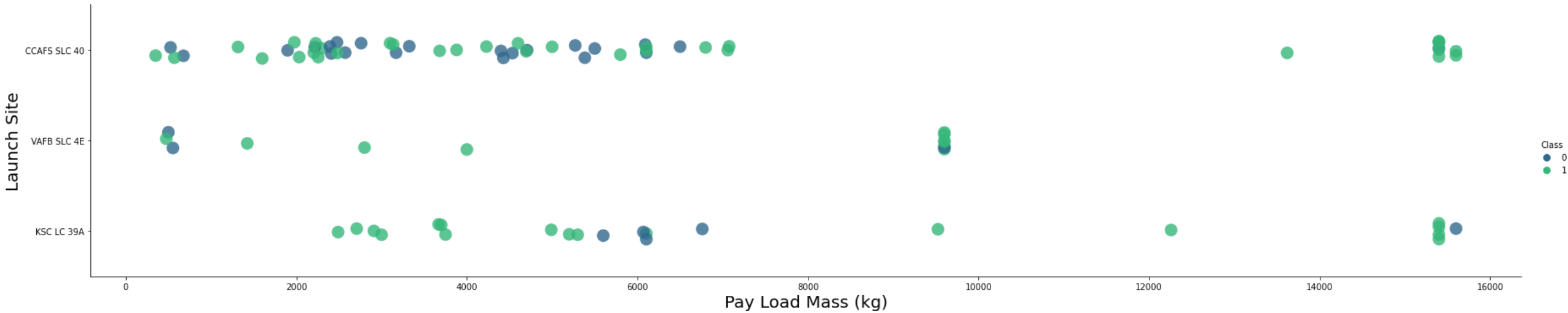
EDA Result 1 — Flight Number vs Launch Site



Green indicates successful launch; Purple indicates unsuccessful launch.

Key finding: Later flight numbers generally show more successful landings, suggesting improvement through operational experience.

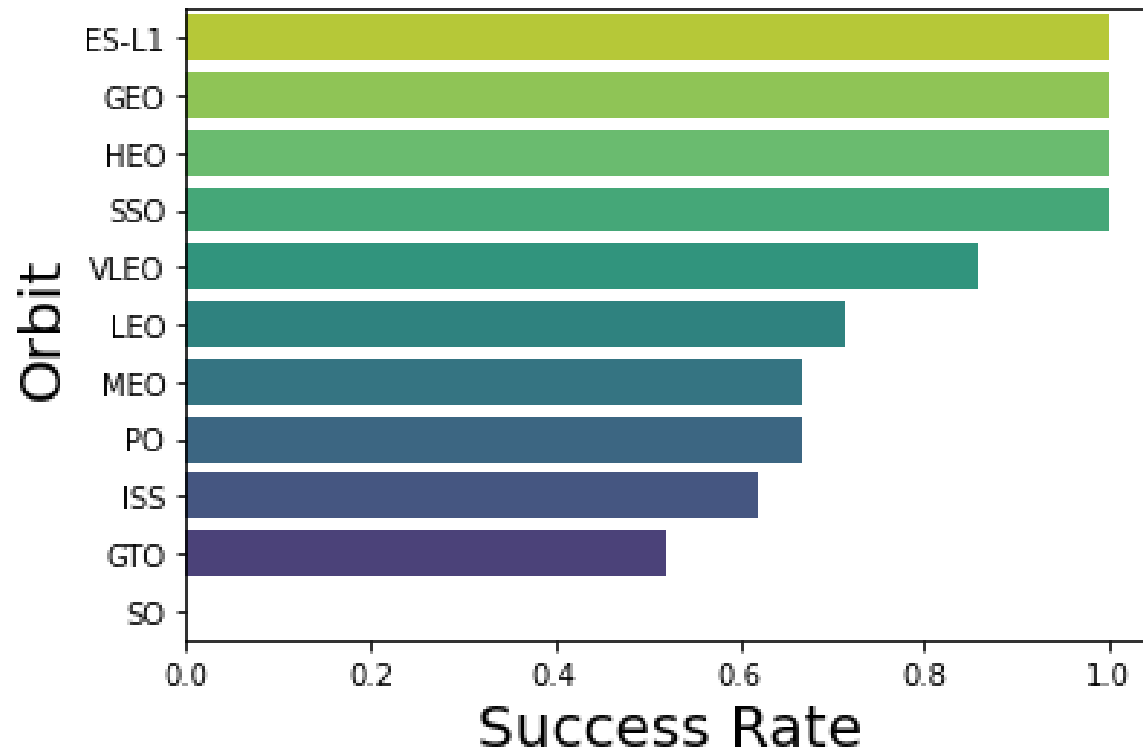
EDA Result 2 — Payload Mass vs Launch Site



Green indicates successful launch; Purple indicates unsuccessful launch.

Key finding: Payload ranges differed across launch sites, but payload mass alone did not fully determine landing success.

EDA Result 3 — Landing Success Rate by Orbit



Key findings: Success rates varied by orbit type, showing that mission requirements may influence booster recovery difficulty.

ES-L1 (1), GEO (1), HEO (1) have 100% success rate (sample sizes in parenthesis)

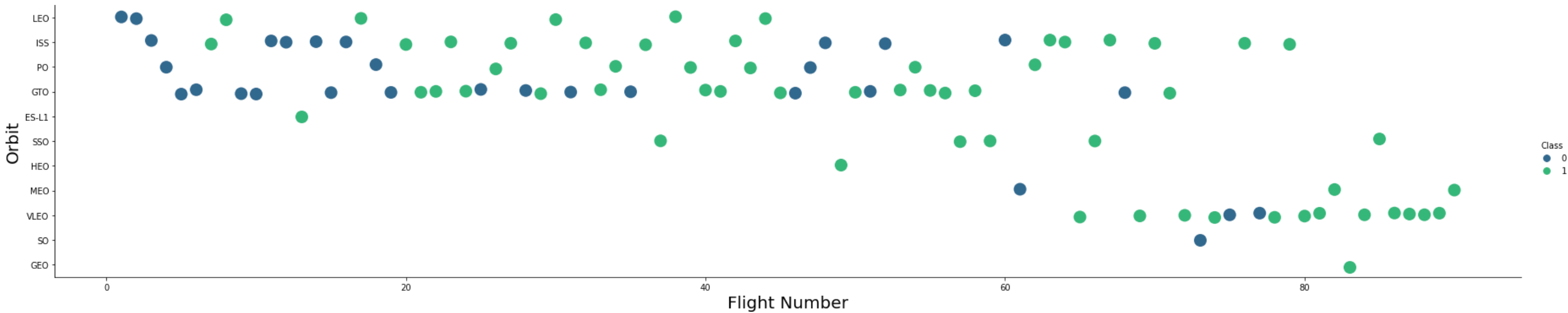
SSO (5) has 100% success rate

VLEO (14) has decent success rate and attempts

SO (1) has 0% success rate

GTO (27) has the around 50% success rate but largest sample

EDA Result 4 — Flight Number vs Orbit

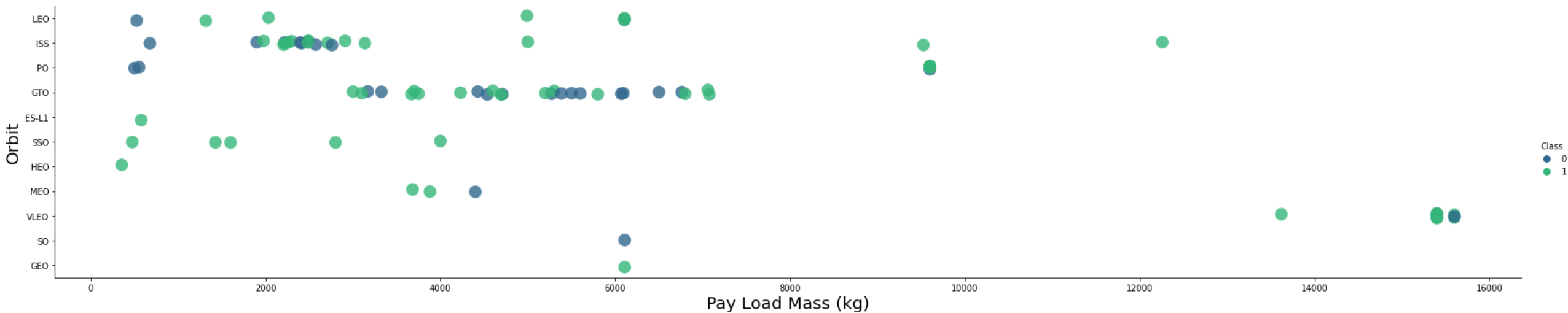


Green indicates successful launch; Purple indicates unsuccessful launch.

Launch Orbit preferences changed over Flight Number.
Launch Outcome seems to correlate with this preference. SpaceX started with LEO orbits which saw moderate success LEO and returned to VLEO in recent launches SpaceX appears to perform better in lower orbits or Sun-synchronous orbits

Key Finding:
Orbit preferences changed over time, while later missions generally showed stronger landing outcomes.

EDA Result 5 — Payload Mass vs Orbit



Green indicates successful launch; Purple indicates unsuccessful launch.

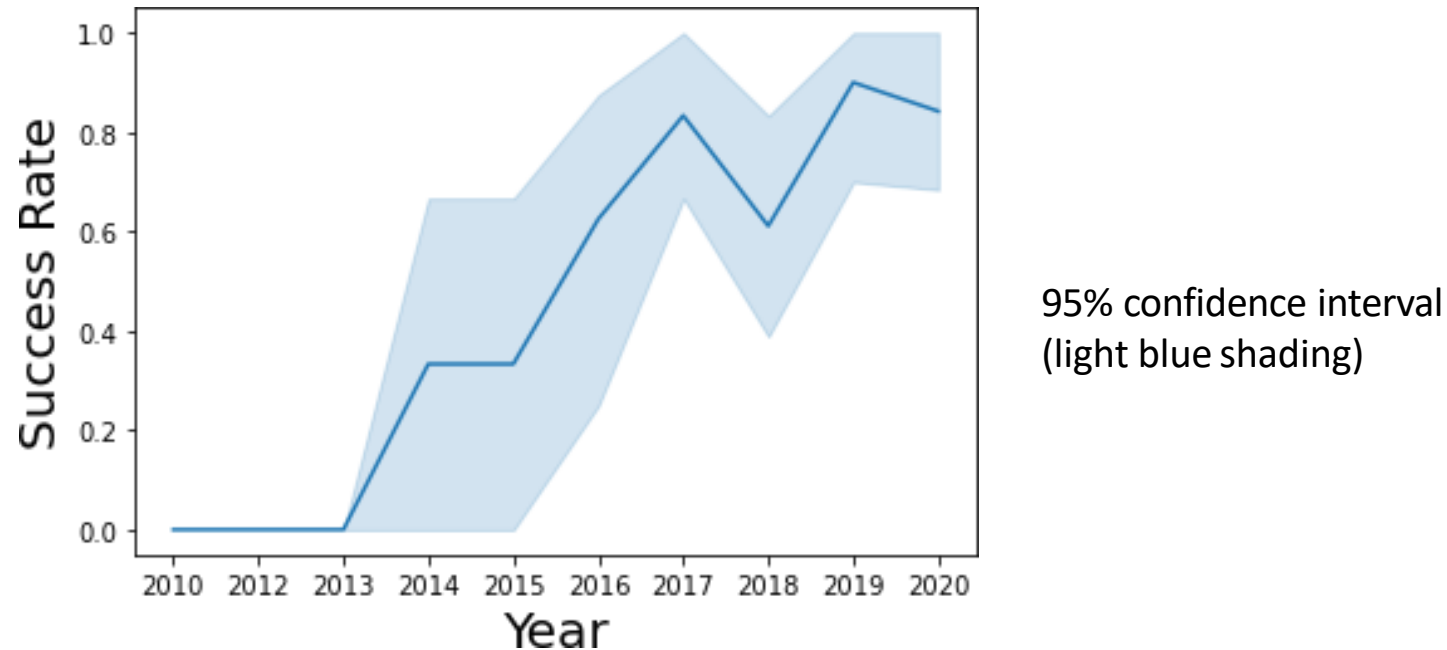
Payload mass seems to correlate with orbit

LEO and SSO seem to have relatively low payload mass

The other most successful orbit VLEO only has payload mass values in the higher end of the range

Key Finding: Payload mass varied by orbit type, showing that mission profile and payload requirements are related.

EDA Result 6 — Yearly Landing-Success Trend



Success generally increases over time since 2013 with a slight dip in 2018

Success in recent years at around 80%

Key Finding: Landing success generally increased over time, with recent years achieving much higher success rates than early missions.

SQL Result 1 — Unique Launch Sites

```
In [4]: %%sql
        SELECT UNIQUE LAUNCH_SITE
        FROM SPACEXDATASET;

* ibm_db_sa://ftb12020:***@0c77d6f:
Done.
```

```
Out[4]:
```

launch_site
CCAFS LC-40
CCAFS SLC-40
CCAFSSLC-40
KSC LC-39A
VAFB SLC-4E

The query returned four recorded launch-site names:

- CCAFS LC-40
- CCAFS SLC-40
- KSC LC-39A
- VAFB SLC-4E

CCAFS LC-40 and CCAFS SLC-40 refer to the same Cape Canaveral location under different names.

Key Finding:

The dataset contains four recorded names representing three physical launch locations.

SQL Result 2 — Launch Sites Beginning with CCA

```
In [5]: %%sql
SELECT *
FROM SPACEXDATASET
WHERE LAUNCH_SITE LIKE 'CCA%'
LIMIT 5;

* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90108kqb1od8l1cg.databases.appdomain.cloud:31198/bludb
Done.
```

Out[5]:

DATE	time__utc__	booster_version	launch_site	payload	payload_mass__kg__	orbit	customer	mission_outcome	landing__outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	07:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	00:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

This query displayed the first five launch-site records whose names begin with “CCA”.

Key Finding:

The matching records are launches from the Cape Canaveral launch site.

SQL Result 3 — Total NASA CRS Payload Mass

```
%%sql
SELECT SUM(PAYLOAD_MASS__KG_) AS SUM_PAYLOAD_MASS_KG
FROM SPACEXDATASET
WHERE CUSTOMER = 'NASA (CRS)';

* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86
Done.
```

sum_payload_mass_kg
45596

This query calculated the total payload mass carried for NASA Commercial Resupply Services missions.

Total Payload Mass:
[45596] kg

Key Finding:
NASA CRS missions carried a substantial combined payload to support International Space Station resupply.

SQL Result 4 — Average Payload Mass for F9 v1.1

```
%%sql
SELECT AVG(PAYLOAD_MASS_KG_) AS AVG_PAYLOAD_MASS_KG
FROM SPACEXDATASET
WHERE booster_version = 'F9 v1.1'
```

```
* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86
Done.
```

avg_payload_mass_kg
2928

This query calculated the average payload mass for launches using the F9 v1.1 booster version.

Average Payload Mass:
2928 kg

Key Finding:

The average F9 v1.1 payload mass was relatively low compared with the maximum payload in the dataset.

SQL Result 5 — First Successful Ground-Pad Landing

```
%%sql
SELECT MIN(DATE) AS FIRST_SUCCESS
FROM SPACEXDATASET
WHERE landing__outcome = 'Success (ground pad)';

* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81
Done.
```

first_success

2015-12-22

This query identified the earliest successful ground-pad landing.

First Successful Ground-Pad Landing:
2015-12-22

Key Finding:

SpaceX achieved its first successful ground-pad landing in December 2015.

SQL Result 6 — Successful Drone-Ship Landings

```
%%sql
SELECT booster_version
FROM SPACEXDATASET
WHERE landing_outcome = 'Success (drone ship)' AND payload_mass__kg_ BETWEEN 4001 AND 5999;

* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8lcg.database
Done.
```

booster_version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

This query returned booster versions that achieved successful drone-ship landings with payload masses greater than 4,000 kg and less than 6,000 kg.

Result:

Four booster versions met the conditions.

Key Finding:

Several Falcon 9 booster versions successfully landed on drone ships while carrying medium-range payloads.

SQL Result 7 — Mission Outcome Counts

```
%%sql
SELECT mission_outcome, COUNT(*) AS no_outcome
FROM SPACEXDATASET
GROUP BY mission_outcome;
```

```
* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-1
Done.
```

mission_outcome	no_outcome
Failure (in flight)	1
Success	99
Success (payload status unclear)	1

This query counted the number of launches for each mission outcome.

Key Finding:

The large majority of launches achieved a successful primary mission outcome, while only a small number had unsuccessful or unclear outcomes.

SQL Result 8 — Maximum Payload Booster Versions

```
%%sql
SELECT booster_version, PAYLOAD_MASS__KG_
FROM SPACEXDATASET
WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXDATASET);

* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1
Done.
```

booster_version	payload_mass__kg_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

This query identified the booster versions that carried the maximum payload mass.

Maximum Payload Mass:
15,600 kg

Key Finding:
The maximum payload was carried by Falcon 9 Block 5 booster variants.

SQL Result 9 — Failed Drone-Ship Landings in 2015

```
%%sql
SELECT MONTHNAME(DATE) AS MONTH, landing__outcome, booster_version, PAYLOAD_MASS__KG_, launch_site
FROM SPACEXDATASET
WHERE landing__outcome = 'Failure (drone ship)' AND YEAR(DATE) = 2015;

* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8lcg.databases.app
Done.
```

MONTH	landing__outcome	booster_version	payload_mass__kg_	launch_site
January	Failure (drone ship)	F9 v1.1 B1012	2395	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	1898	CCAFS LC-40

This query returned 2015 launch records where the first stage failed to land on a drone ship.

Result:

Two records met the conditions.

Key Finding:

Two failed drone-ship landing attempts were recorded in 2015.

SQL Result 10 — Ranked Landing Outcomes by Date

```
%%sql
SELECT landing__outcome, COUNT(*) AS no_outcome
FROM SPACEXDATASET
WHERE landing__outcome LIKE 'Succes%' AND DATE BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY landing__outcome
ORDER BY no_outcome DESC;
```

```
* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8lce
Done.
```

landing__outcome	no_outcome
Success (drone ship)	5
Success (ground pad)	3

This query ranked successful landing outcomes between 4 June 2010 and 20 March 2017.

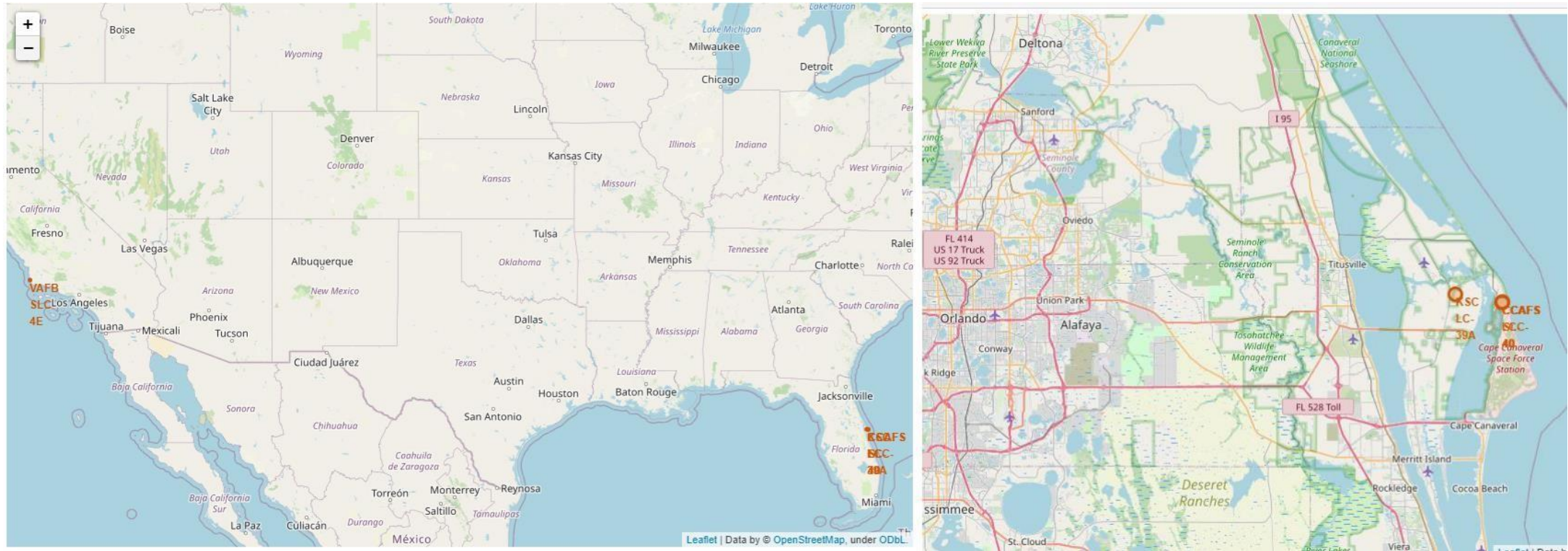
Result:

- 8 successful landings in total
- Successful outcomes included drone-ship and ground-pad landings

Key Finding:

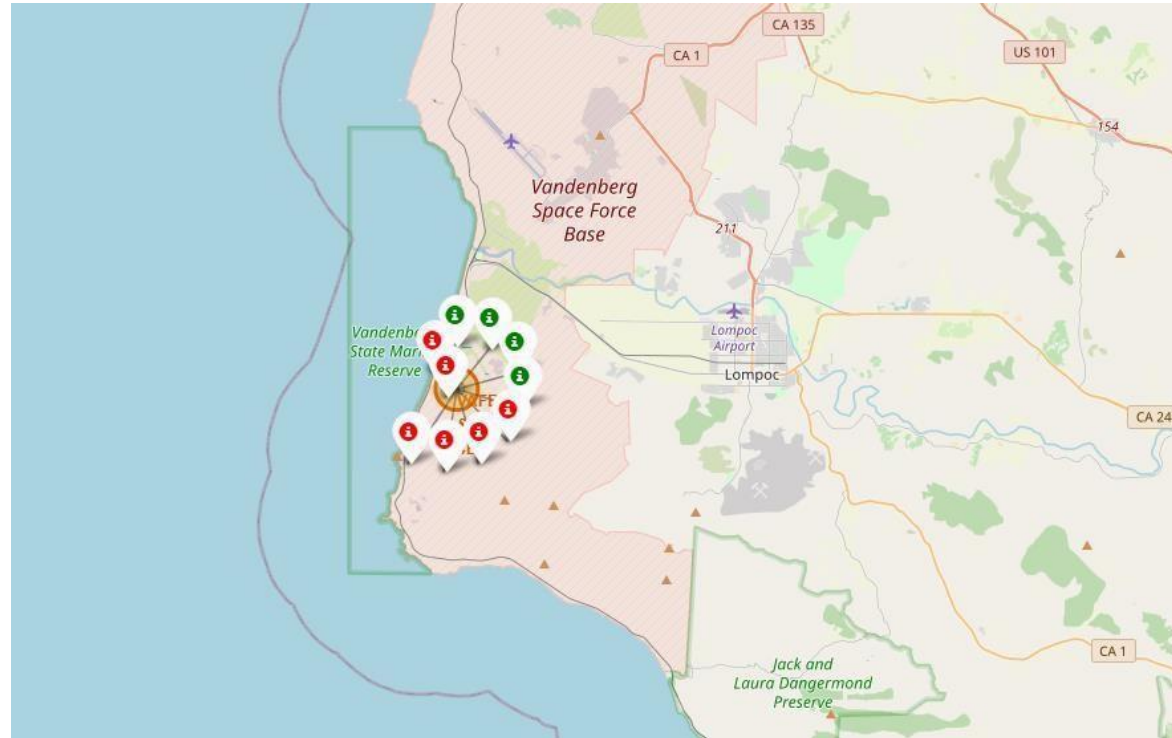
Eight successful landings occurred during the selected period.

Folium Result 1 — Launch-Site Locations



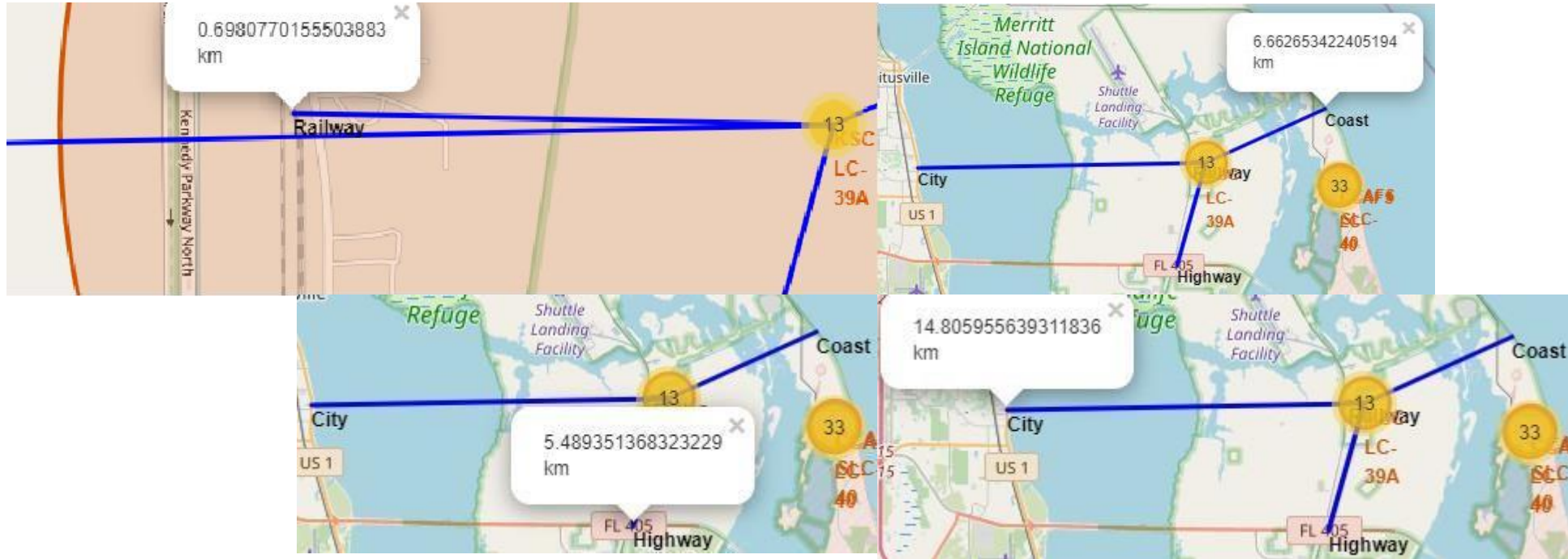
The left map shows all launch sites relative US map. The right map shows the two Florida launch sites since they are very close to each other. All launch sites are near the ocean.

Folium Result 2 — Successful and Failed Landing Markers



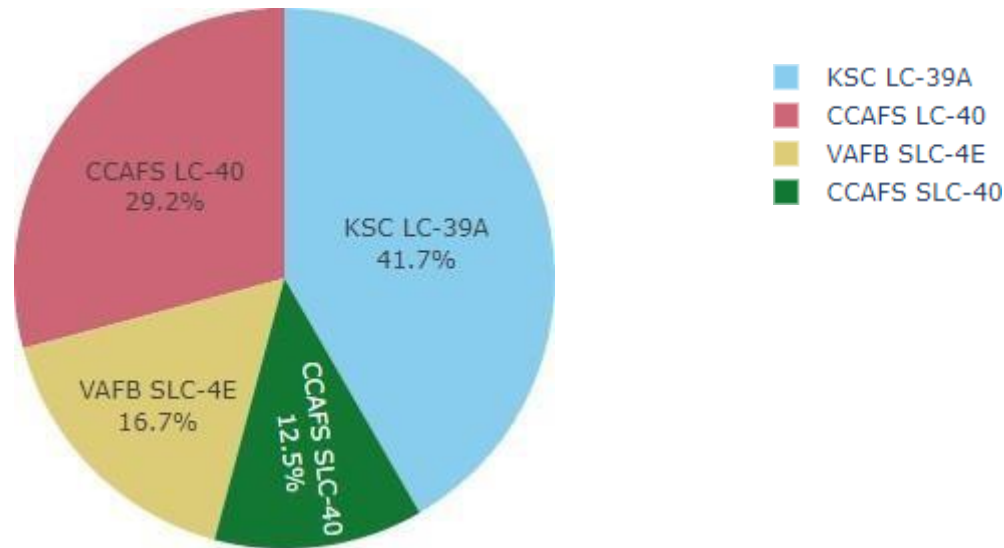
Clusters on Folium map can be clicked on to display each successful landing (green icon) and failed landing (red icon). In this example VAFB SLC-4E shows 4 successful landings and 6 failed landings.

Folium Result 3 — Launch-Site Proximity Analysis



Using KSC LC-39A as an example, launch sites are very close to railways for large part and supply transportation. Launch sites are close to highways for human and supply transport. Launch sites are also close to coasts and relatively far from cities so that launch failures can land in the sea to avoid rockets falling on densely populated areas.

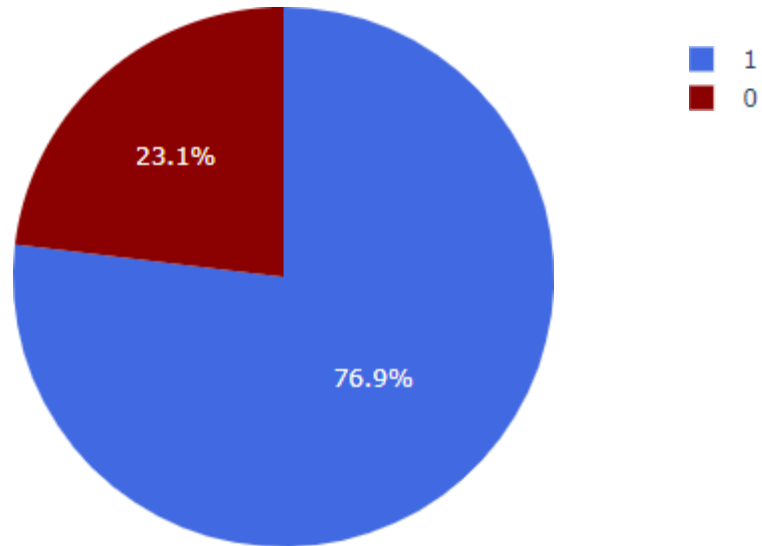
Plotly Dash Result 1 — Successful Landings by Launch Site



This is the distribution of successful landings across all launch sites. CCAFS LC-40 is the old name of CCAFS SLC-40 so CCAFS and KSC have the same amount of successful landings, but a majority of the successful landings were performed before the name change. VAFB has the smallest share of successful landings. This may be due to smaller sample and increase in difficulty of launching in the west coast.

Plotly Dash Result 2 — Launch Site with Highest Success Rate

KSC LC-39A Success Rate (blue=success)



KSC LC-39A has the highest success rate with 10 successful landings and 3 failed landings.

Plotly Dash Result 3 — Payload Mass vs Landing Outcome

Payload range (Kg):

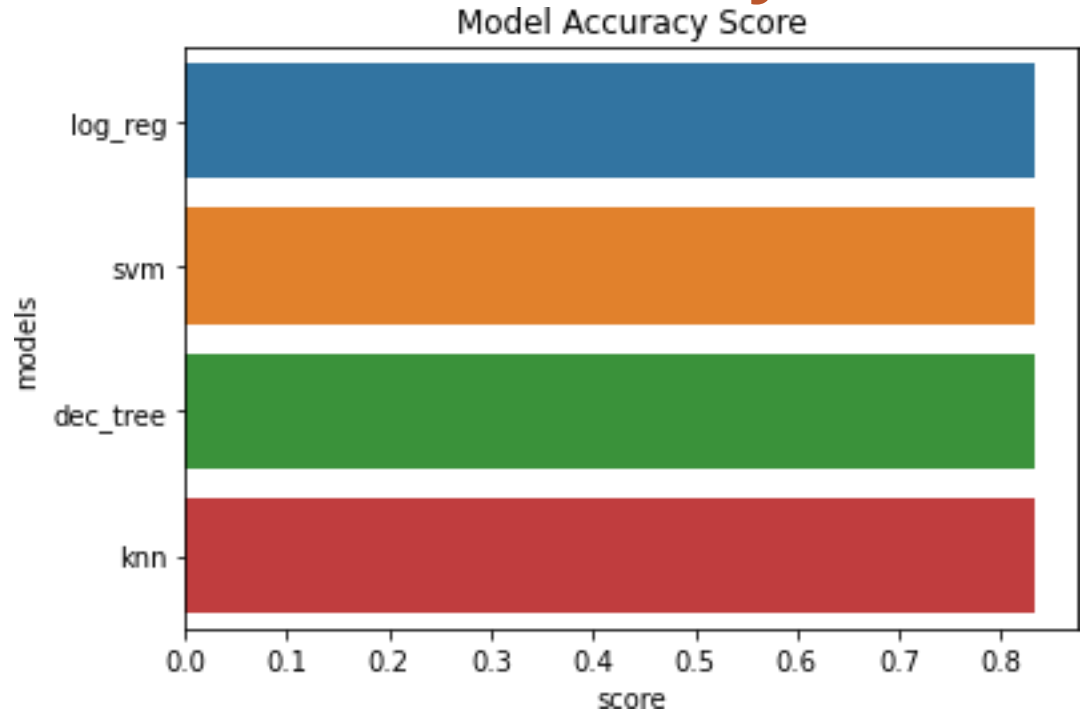


Payload Mass vs. Success vs. Booster Version Category



Plotly dashboard has a Payload range selector. However, this is set from 0-10000 instead of the max Payload of 15600. Class indicates 1 for successful landing and 0 for failure. Scatter plot also accounts for booster version category in color and number of launches in point size. In this particular range of 0-6000, interestingly there are two failed landings with payloads of zero kg.

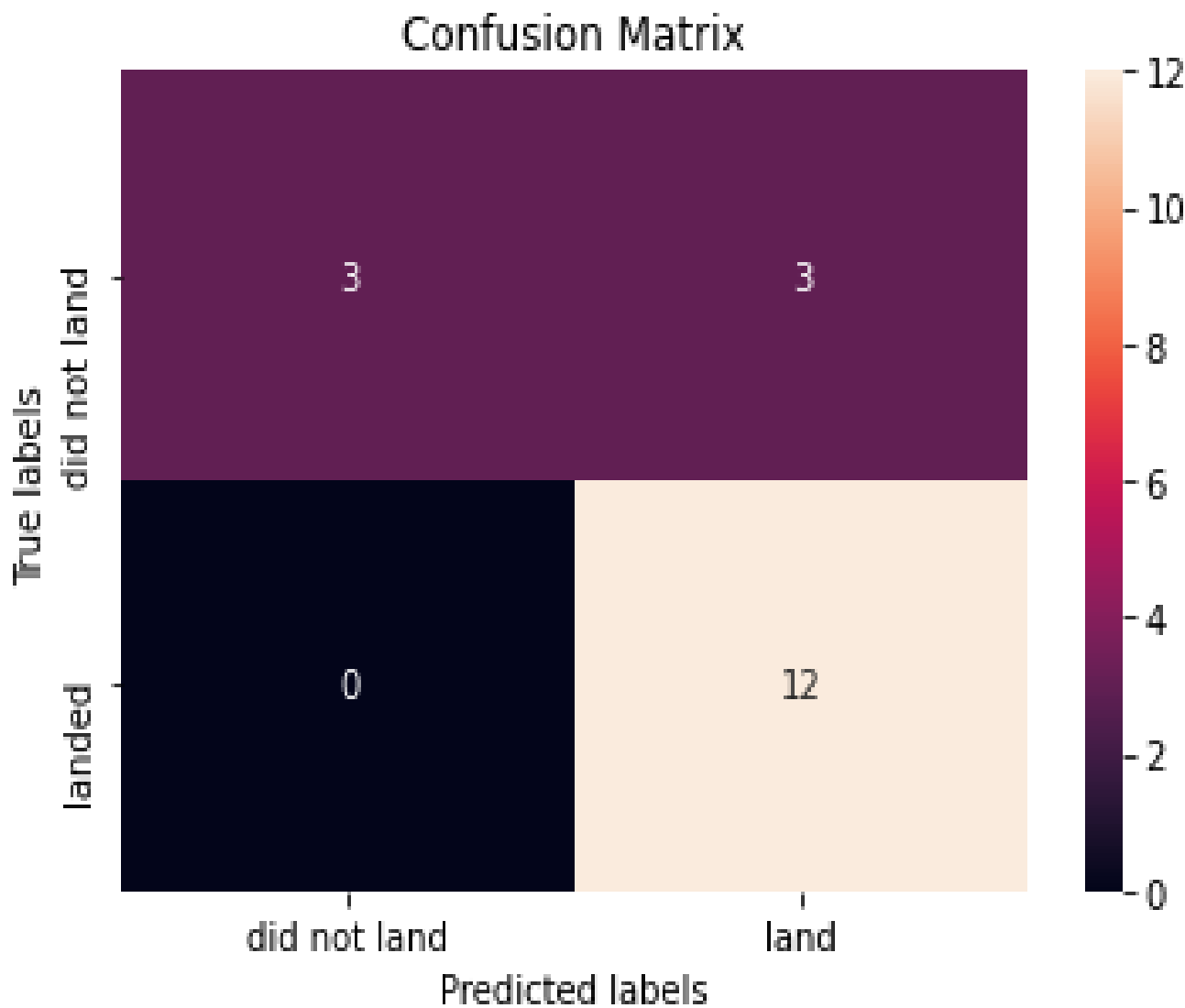
Predictive Analysis Results — Model Comparison



Model	Test Accuracy
Logistic Regression	83.33%
Support Vector Machine	83.33%
Decision Tree	83.33%
K-Nearest Neighbours	83.33%

Key Finding:
All four models achieved similar test accuracy. Because the test set contained only 18 records, more data is needed to confidently identify the strongest model.

Predictive Analysis Results — Confusion Matrix



- 12 successful landings were predicted correctly.
- 3 unsuccessful landings were predicted correctly.
- 3 unsuccessful landings were incorrectly predicted as successful.
- The models therefore tended to overpredict successful landings.

Key Finding:

Accuracy alone does not show the type of error; the confusion matrix shows that false positives were the main prediction issue.

Innovative Insights and Recommendations

- Flight number can act as a measure of operational experience, since later missions generally achieved higher landing success.
- Orbit and payload requirements may affect the fuel remaining for booster recovery.
- Coastal launch-site placement reduces public risk, while nearby highways and railways support logistics.
- Similar accuracy across all four models suggests that the limited dataset may matter more than model complexity.
- Future models could include weather, sea conditions, booster age, refurbishment history, and fuel-reserve information.

CONCLUSION

- • This project applied a complete data science workflow to SpaceX Falcon 9 launch data.
- • Data was collected using the SpaceX API and web scraping, cleaned using Pandas, and analysed using Python visualizations and SQL.
- • Folium maps showed that launch sites are located near coastlines and transportation infrastructure.
- • The Plotly Dash dashboard enabled interactive analysis of launch-site success rates and payload relationships.
- • Landing success generally improved over time and varied by launch site, orbit, payload mass, booster version, and flight experience.
- • Logistic Regression, SVM, Decision Tree, and KNN achieved approximately 83.33% test accuracy.
- • Because the test set contained only 18 records, additional data is required for stronger model comparison and more reliable predictions.

GitHub repository url:

<https://github.com/hodzz/IBM-SpaceX-Data-Science-Capstone>